

AN OPEN SOURCE SOFTWARE PACKAGE TO COMPARE AND EXTEND EFFECTIVE MASS MODELS OF THE PHOSPHORUS DONOR IN SILICON

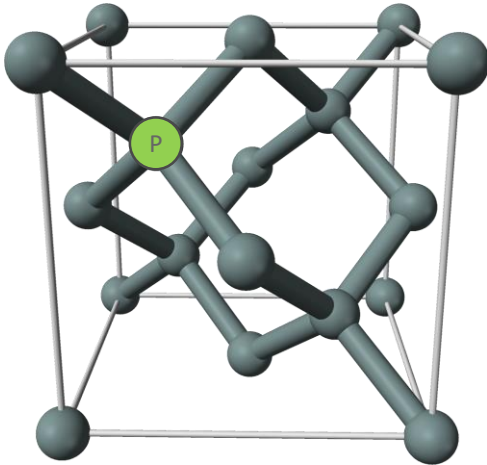
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Presentation Date: Monday, March 6, 2023 4:36 pm

gitlab.com/cantankerous-impurity

OUR PROBLEM

System description



- Infinite silicon crystal bulk.
- One silicon replaced by an isolated pnictogen (Series V/Group 15).
- Specifically we consider the **phosphorous** substitutional donor.
- System is symmetric about the impurity under **operations of point group T_d** .
- Excess electric charge of phosphorous nuclear attracts excess electron.

Original crystal structure picture by Ben Mills, public domain.

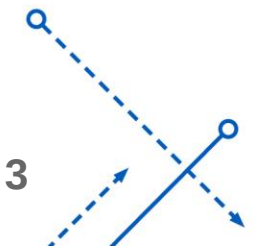
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Modified by L. Pendo for presentation.

The back of the envelope idea

- Excess electron is affected by a **periodic mean field potential**, and **weakly bound to donor potential**.
 - **Short-range**: looks like a **Bloch wave**.
 - **Long-range**: **attenuated on a radial scale** much longer than lattice parameter.

[https://gitlab.com/cantankerous-impurity/Documentation/-/tree/core/Presentations/MarchMeeting2023\(LasVegas\)](https://gitlab.com/cantankerous-impurity/Documentation/-/tree/core/Presentations/MarchMeeting2023(LasVegas))



THE MODEL

The effective mass approximation

EMA: Assume the donor electron is localized in $\mathbf{k}$ -space at the **six-fold degenerate minima** of the silicon conduction band.

$$|\Psi\rangle = \sum_n \int_{BZ} d\mathbf{k} A_{\mathbf{k}} |n\mathbf{k}\rangle \rightarrow \sum_{\mathbf{k}_p} \beta_{\mathbf{k}_p} \int_{BZ} d\mathbf{k} A_{\mathbf{k}}^{(p)} |0\mathbf{k}\rangle$$

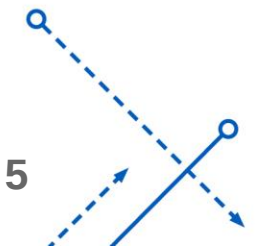
$$\Psi_{\beta;\mathbf{r}} \approx \sum_{\mathbf{k}_p} \beta_{\mathbf{k}_p} F_{\mathbf{r}}^{(p)} \Theta_{\mathbf{r}}^{(p)}$$

$$F_{\mathbf{r}}^{(p)} := \int_{BZ} d\mathbf{k} A_{\mathbf{k}}^{(p)} e^{i(\mathbf{k}-\mathbf{k}_p)\cdot\mathbf{r}} \quad \Theta_{\mathbf{r}}^{(p)} := e^{i\mathbf{k}_p\cdot\mathbf{r}} u_{\mathbf{k}_q\mathbf{r}}$$

- *Short-range:* Bloch wave.
- *Long-range:* Envelope function.

We expand the envelope function with a set of basis functions.

$$F_{\mathbf{r}}^{(q)} = \sum_j c_j \Phi_{j\mathbf{r}}^{(q)}$$



The Generalized Eigenvalue Problem

“Modified” Shindo-Nara effective mass equation[1] cast as a *generalized eigenvalue problem*[2].

$$E\mathcal{S}c = (\mathcal{T} + \mathcal{U})c$$

$$c^\dagger \mathcal{S}c := \langle \Psi_\beta | \Psi_\beta \rangle \quad c^\dagger \mathcal{T}c := \langle \Psi_\beta | \hat{H}_0 | \Psi_\beta \rangle \quad c^\dagger \mathcal{U}c := \langle \Psi_\beta | \hat{U} | \Psi_\beta \rangle$$

H_0 is intended to model the pure silicon potential.

U is intended to model *all* deviations from the pure silicon potential.

symmetrized anisotropic basis states

$$\Psi_{\beta\mathbf{r}} \approx \sum_j c_j \left(\sum_{q \in \{x,y,z\}} \beta_q \Phi_{j\mathbf{r}}^{(q)} \Theta_{\Omega\mathbf{r}}^{(q)} \right)$$

Bloch functions of *opposed valley minima* are included as Fourier series: $\Theta_{\Omega\mathbf{r}}^{(q)} := \frac{\Theta_{\mathbf{r}}^{(q)} + \Omega \Theta_{\mathbf{r}}^{(-q)}}{\sqrt{2}}$



Views on the effective mass approximation

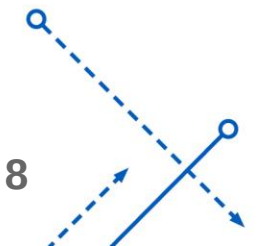


- An attractively intuitive physical picture of the state of the donor electron.
- Less computationally expensive as compared to *ab initio* and tight-binding models.
- Predicts the energies of spatially extended excited states of donors [1,2].



- Struggles to present a quantitatively persuasive picture of the ground state manifold
- Few agreed upon criteria for refutability (beyond agreement with energy spectrum).
- Lots of freedom to generate different schemes through reasonable but ad hoc assumptions.

Is further improvement of effective mass methods possible?



Model evaluation

Out of model

Verisimilitude

How well does the model resemble reality?

In-model

Consistency

How well does the model play by its own rules?

Completeness

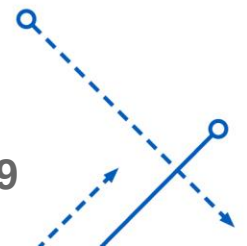
Can the model answer questions posed to it?

V-score[*]: Metric for model completeness

Is the trial eigenstate **well-converged** to an actual eigenstate?

$$v^2 := \frac{\langle \hat{H}^2 \rangle}{E^2} - 1$$

$$\langle \hat{H}^2 \rangle := \frac{\langle \Psi_\beta | \hat{H}_0^2 + \hat{H}_0 \hat{U} + \hat{U} \hat{H}_0 + \hat{U}^2 | \Psi_\beta \rangle}{\langle \Psi_\beta | \Psi_\beta \rangle}$$



THE BASIS

Complete, Extensible Basis

Representation of single eigenstate

- Underlying naïve subbases are *orthonormal*, *fully complete* over $L^2(\mathbb{R}^3)$.
- Two *variational* parameters: *anisotropy* along azimuth, overall *radial* scaling.
- **Users can select membership and size of each *subbasis*.**
- However, difficult to optimize *subbasis* to span over multiple energy eigenstates.

Representation of multiple eigenstates / energy spectra

- *Superbasis* concatenation of individual independent *subbases*.
- *Superbasis* *non-orthonormal*, *overcomplete* on $L^2(\mathbb{R}^3)$.
- Optimization of each *subbasis* seems to mitigate problems from overcompleteness.
- **Total number and composition of subbases limited ONLY by computation limits.**
- Work on-going to increase code efficiency to allow for larger sub- and super-bases.

POTENTIAL MODEL

Flexible Potential Model

To facilitate deep exploration, **a very general potential model**.

$$U_{\mathbf{r}} = \sum_i \mathcal{U}_i r^{\epsilon_i} \sum_j \mathcal{U}_{ij\{\Omega\}} e^{-\mathcal{A}_{ij\{\Omega\}} r} e^{-\mathcal{B}_{ij\{\Omega\}} r^2}$$

$$\mathcal{F}_{ij\{\Omega\}} = f_{ij,0} + f_{ij,x} \chi_x + f_{ij,y} \chi_y + f_{ij,z} \chi_z$$

$$\chi_x = \frac{x}{r} = \sin(\theta) \cos(\phi) \quad \chi_y = \frac{y}{r} = \sin(\theta) \sin(\phi) \quad \chi_z = \frac{z}{r} = \cos(\theta)$$

- The fraktur constants can be freely set by users of the software.
- Potential form **represents most extant models** of the effective silicon donor potential.
- Examples include the tetrahedral potential used by Gamble [1] and the dynamic dielectric model [2,3].

$$U_{\text{Gamble}} = -\frac{2}{r} + A_0 e^{-\beta_0 r^2} + A_1 \sum_{i=1}^4 e^{-\beta_1 |\mathbf{r} - b \mathbf{t}_i|^2}$$

$$\mathbf{t}_i = \{[111], [\bar{1}1\bar{1}], [1\bar{1}\bar{1}], [\bar{1}\bar{1}1]\}$$

$$U_{\text{Dyn}} = -\frac{2}{r} (1 + A e^{-\alpha r} + (\epsilon_0 - A) e^{-\beta r} - e^{-\gamma r})$$

TEST CASE

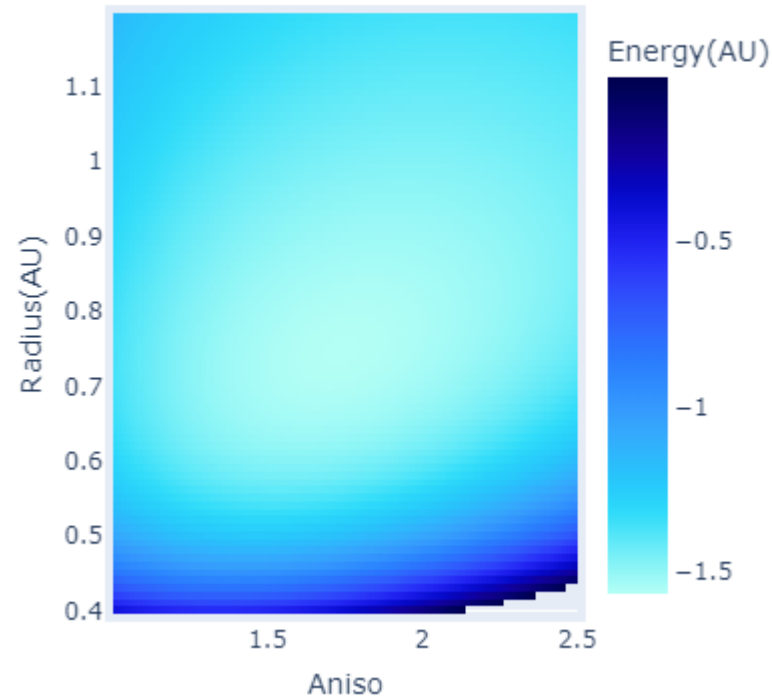
*Single-Valley Wavefunctions
Beyond Kohn-Luttinger*

Kohn-Luttinger (Historic)

$$\epsilon_{AU} = 19.944 \text{ meV}$$

$$a_{AU} = 3.167 \text{ nm}$$

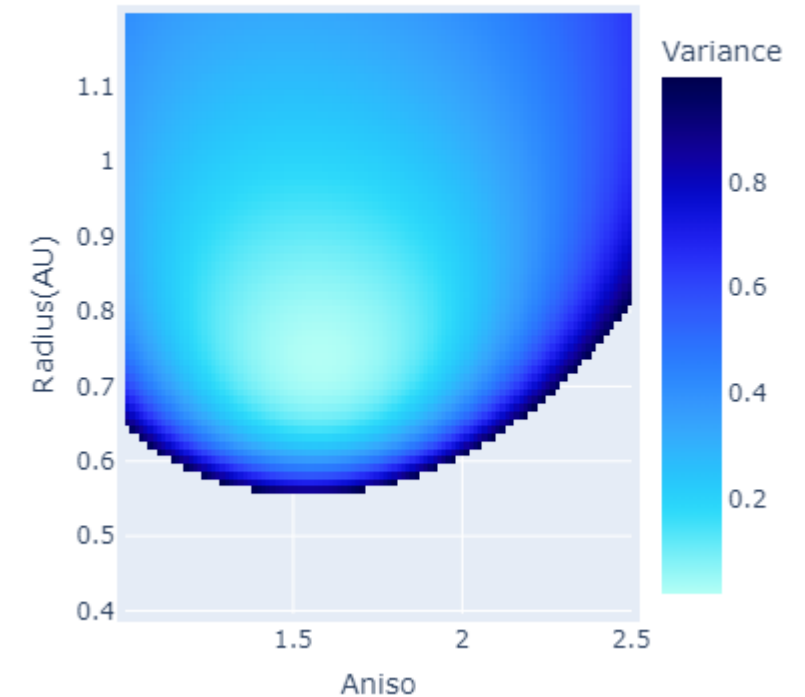
Basis(m,l,k)
0,0,0



$$\lambda \rightarrow 1.745 \quad \eta \rightarrow 0.7522$$

$$E[\lambda, \eta] = -1.565 \quad v[\lambda, \eta] = 0.05048$$

$$E[\lambda, \eta] = -31.22 \text{ meV}$$



$$\lambda \rightarrow 1.585 \quad \eta \rightarrow 0.7421$$

$$E[\lambda, \eta] = -1.558 \quad v[\lambda, \eta] = 0.01877$$

$$E[\lambda, \eta] = -31.08 \text{ meV}$$

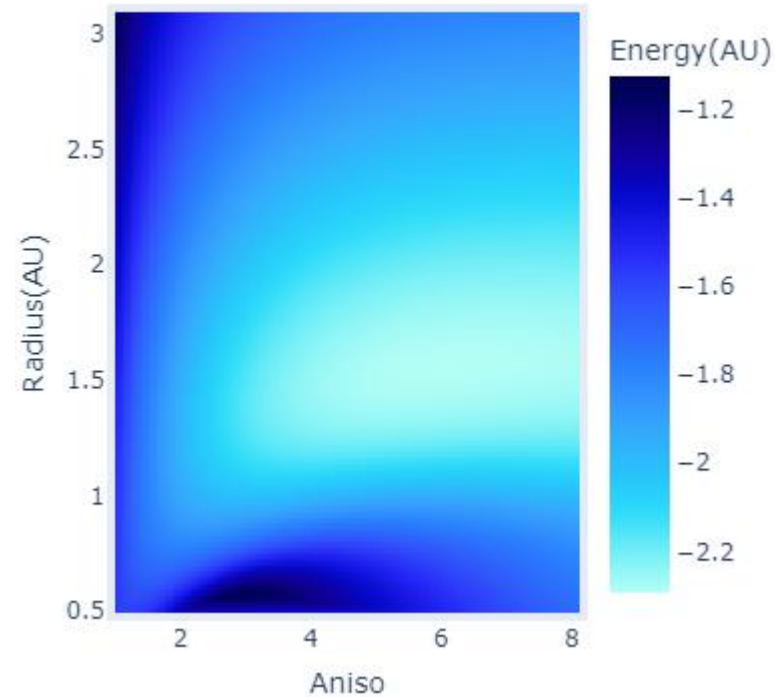
Expanded Single-Valley

$$\mathfrak{E}_{AU} = 19.944 \text{ meV}$$

$$\mathfrak{a}_{AU} = 3.167 \text{ nm}$$

Basis(m,l,k)

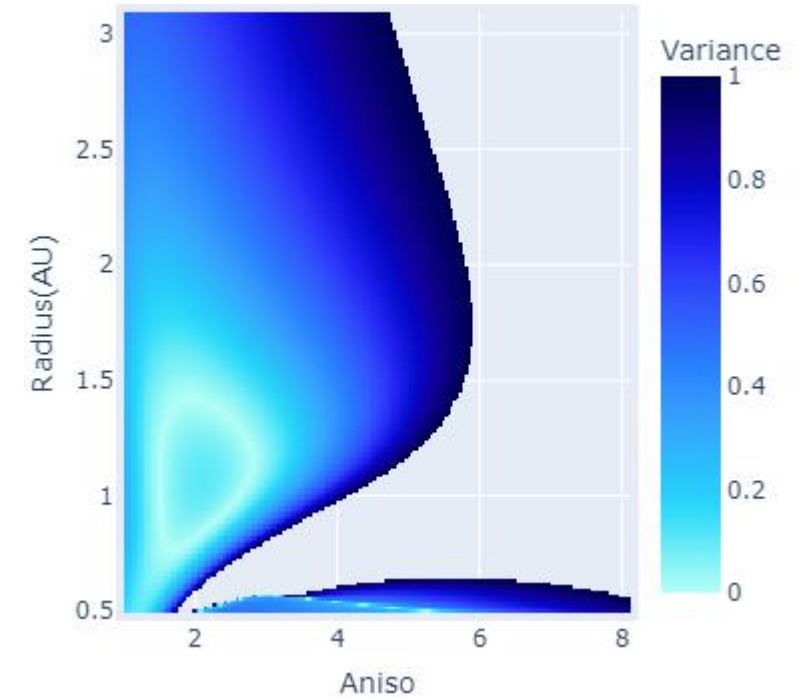
0,0,0
0,0,1
0,0,2
0,0,3
0,2,0
0,2,1
0,2,2
0,2,3
0,4,0
0,4,1
0,4,2
0,4,3
0,6,0
0,6,1
0,6,2
0,6,3



$$\lambda \rightarrow 6.130 \quad \eta \rightarrow 1.558$$

$$E[\lambda, \eta] = -2.294 \quad v[\lambda, \eta] = 1.092$$

$$E[\lambda, \eta] = -45.74 \text{ meV}$$



$$\lambda \rightarrow 1.628 \quad \eta \rightarrow 1.349$$

$$E[\lambda, \eta] = -1.832 \quad v[\lambda, \eta] = 0.00001905$$

$$E[\lambda, \eta] = -36.54 \text{ meV}$$

CONCLUSIONS

Prospects

- Specific focus of my PhD research on Si:P and the EMA, but *framework is very general*.
- Want to make code and documentation available as an open science, open notes, open source GPL 3 licensed project.

Can be used to explore other Hamiltonians, other symmetries, other systems.

Extensions

- Perturbations of T_d symmetry.
- External fields, electric, magnetic.
- Multi-center effects.
- Multi-electron effects.

Forks

- Different donors in silicon.
- Different bulk materials.
- 2-D systems: quantum dots.

“The Pitch”

Upshot: Want others to benefit from 10,000+ lines of code, thousands of work hours.

- **Opportunity for a collaboration led by PhD students, postdocs, early career scientists.**
- **A chance to develop scientific, technical, logistic, communication, and management skills.**
 - **Python:** Symbolic computations, data analysis, data visualization, SQL integration.
 - **C++:** Code analysis, optimization, documentation. Work with new features of latest C++ standard.
 - **Fortran:** Integration of scientific code from multiple languages.
 - **Mathematics:** Symmetry group theory. Numerical stability. Stochastic optimization. Function representation.

Interested?

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- Gitlab repo: cantankerous-impurity